Global Reporting Regulation, Stock Liquidity, and Policy Diffusion

Abstract

We examine stock liquidity around 17 families of reporting regulation and their international diffusion. The issuer panel contains 1,055,142 security-years in 90 jurisdictions. Geographic exposure contrasts yield first-year log-spread estimates of -0.049 with equal family weights and +0.015 with market-capitalisation family weights. A separate source-corroborated analysis covers six families: auditor reporting, digital filing, conflict minerals, banking disclosure, IFRS and statutory nonfinancial reporting. Canadian bank IFRS releases and German first-cycle statutory reports extend this evidence using local archives and primary issuer sources. Across these selected studies, first-year log-spread estimates are -0.090 and +0.191 under the two weighting schemes. Longer-window support covers four source-supported families. Other outcomes, pretrends, prior voluntary disclosure and common legal waves limit a general causal liquidity interpretation. No diffusion coefficient survives broad multiplicity adjustment. The findings show why legal scope, information timing and weights matter when combining reporting policies.

1. Introduction

Disclosure regulation can improve liquidity by supplying credible information to investors. Whether this channel operates depends on what is disclosed, who receives it, and which firms are required to report. A confidential tax return, a public auditor report and a crypto-asset white paper do not impose the same information shock on ordinary equity markets. We ask how liquidity changes around these rules and whether prior foreign implementation helps explain subsequent policy choices.

We follow the empirical logic of Krueger et al. (2023): establish the reporting obligation, connect it to firms and information timing, examine complementary liquidity outcomes, and assess identification before drawing conclusions. Our setting extends beyond comprehensive sustainability reporting to 17 economic regulation families. We then use the competition, learning, emulation and coercion framework in Koning, Mertens and Roosenboom (2018) to examine diffusion.

The analysis distinguishes two populations. The broad panel assigns geographic exposure to individual vendor-issuer outcomes on country-family calendars. Source-corroborated studies instead require specific legal, filing or report evidence. Their six families include IFRS transition reporting and first statutory nonfinancial statements. This permits a broad comparison without implying that every company in a country is legally covered. Banking phases enter each aggregate once.

Table 2 gives equal and market-capitalisation family weights equal prominence. First-year equal-family spreads decline in both populations, while market-capitalisation estimates are less favourable. Complementary outcomes and pre-policy differences constrain interpretation. Table 3 identifies each source-supported study, including small treated groups. The estimates apply to these policy mixes and selected issuer populations; they do not establish a common causal reporting benefit.

2. Institutional setting and research design

Public reporting can reduce adverse selection when it supplies incremental information. Enforcement, prior disclosure and compliance costs can moderate the effect. Comprehensive mandatory E/S/G requires a response on environmental, social and governance topics; the final topic determines the joint date. Comply-or-explain requires a response, while CSR spending and climate-only rules do not establish all-topic disclosure. The inherited review retains 37 qualifying country phases from 41 reviews; earliest separate topic histories remain incomplete. In the reference study, voluntary earnings guidance is an information-environment proxy, rather than a voluntary ESG treatment.

Eligibility is rule specific (Table 1). Confidential CbCR goes to tax authorities; its actual return date is not a public disclosure date. Public bank CbCR attaches to credit institutions and consolidation perimeters, with partial and full phases. Listed holding companies are not automatically the regulated reporter. Beneficial-ownership rules can exempt securities-reporting issuers, while MiCA concerns crypto assets and services. Country calendars therefore cannot certify ordinary-equity coverage.

The 2000-2025 market panel contains 99,100 vendor quote/company identifiers and 1,055,142 security-years. Outcomes are log median relative quoted spread, log median Amihud price impact and the fraction of valid observed returns equal to zero. Each requires 120 valid observations; logged proxies use annual 1/99 winsorisation. The calendar-year and observed-return denominator differ from the reference paper. A vendor code is not a complete historical legal-entity register. Source archives link period-specific reports to named issuers and historical share identifiers.

For each family, cohort and endpoint, we subtract the weighted change from year g -1 among known later implementers from the change among implementers in g. We select one company home quote by pre-event market capitalisation and require fixed issuer, country and outcome coverage. Geographic studies require at least 30 issuers per country and three comparison markets. First-year contrasts use g -1 and g; balanced contrasts fix support from g -3 to g +2 and average years 0-2. Unknown histories are never classified as never-treated.

GDP in g -1 is allocated across sampled issuers within countries. In covered studies, the denominator is the fixed source-study country universe, so each cohort does not receive a fresh country GDP mass. Cohort weights equal treated economic mass. Across families, equal weights assign 1/F; market-cap weights use summed pre-event treated issuer capitalisation. Equal-issuer and issuer-cap within-family alternatives are disclosed. Repeated policy exposures are not a unique-issuer investment portfolio. For IFRS, comparison firms are US domestic bank parents with contemporaneous auditor US GAAP language. For German statutory reports, comparison parents must be US incorporated; separately covered German subsidiaries remain possible. Source candidates are ranked by pre-event capitalisation before new liquidity coefficients are inspected.

Country influence functions preserve covariance across geographic families; 1,999 shared multiplier draws yield conditional intervals, lead diagnostics and Holm adjustment across 12 principal coefficients. Collective legal waves, concurrent reforms and spillovers still limit policy inference. Covered-study intervals describe issuer sampling only. Few common regulatory waves do not support causal policy p-values. Their minimum of three treated and twenty comparison issuers permits sparse case studies, not a representative global census.

3. Liquidity results

Table 2 Panel A reports issuer outcomes under geographic exposure. Fifteen families support the first year. Equal-family log spread is -0.049 and price impact is -0.072; the zero-return fraction falls by 0.00764. Market-cap family estimates are +0.015, +0.009 and -0.00249, with conditional intervals including zero. Confidential tax CbCR receives 33.1% of this exposure weight, although the report audience is tax authorities.

The balanced geographic comparison supports nine families. Equal-family spread is -0.032 and price impact is -0.013; market-cap values are +0.040 and +0.095. Joint lead diagnostics reject for spreads under both schemes. Different horizons change support and cannot be interpreted as one dynamic path. Timing shifts, independent-choice sensitivities and complete event profiles remain available.

Table 3 retains auditor reporting, early XBRL, conflict minerals and observed bank publication studies. CAM first-year spread is +0.236, with 1,481 treated and 2,767 comparison issuer-cohort observations. XBRL supports six treated issuers and 62 later-phase comparisons valid through 2009 only. Conflict minerals and observed full bank reporting each have three source-selected treated issuers; their comparator and consolidation limits remain explicit.

The IFRS study follows Royal Bank of Canada, Toronto-Dominion, Bank of Montreal and Bank of Nova Scotia. Their first required IFRS interim statements cover the quarter ended 31 January 2012 and are public on 28 February,1 March or 6 March 2012. Earlier 2012 restatement releases are recorded separately. Forty-two US banks have first-year outcome support and contemporaneous domestic-parent/US GAAP evidence. The GDP-weighted spread contrast is -0.235, but price impact rises by 0.132. Balanced support contains four treated banks and 39 comparisons. Canadian-US banking differences and anticipation limit identification.

The statutory nonfinancial study follows BASF, Bayer and BMW. Original 2017 reports recovered from the all-data EU archive explicitly identify German HGB nonfinancial obligations. Bayer and BMW describe a first statutory 2017 cycle; prior voluntary reports are observed. Public report year 2018 is corroborated. BASF and Bayer report/results bundle dates are 27 and 28 February 2018; BMW retains year precision because a scheduled March date does not by itself establish actual publication. Three treated issuers and 35 US parent comparisons support first-year spread -0.217, price impact -0.235 and zero returns -0.01242. Balanced support retains 31 comparisons. Monsanto/Covestro changes at Bayer, industry composition, European reforms and subsidiary spillovers are material competing explanations.

Panel B combines these six source-supported families, counting banking once. First-year equal-family spread is -0.090 and market-cap spread is +0.191. Price-impact estimates are +0.002 and +0.190; zero-return fractions rise under both schemes. Figure 1 shows every supported family. The four-family balanced subset yields spreads of -0.073 and -0.081, while price impact rises. Figure 2 shows material pre-policy differences in spreads and price impact. Sparse selected issuers, common legal waves and changing family support preclude an all 17-family covered-issuer causal interpretation.

4. International diffusion and conclusion

Following Koning et al., competition uses prior implementation among export-pattern peers; learning uses prior outcomes among geographic neighbours; emulation uses language, legal-origin and IGO peers; external dependence uses IMF credit and aid. Signals precede the focal decision. Family-year strata and country-clustered covariance accompany timing-logit and status alternatives, with lagged income, growth and trade controls where specified. The ledger identifies first recorded phases, not certified first-ever adoptions.

Across 1,330 attempted designs, 433 stable focal coefficients remain; none passes the broad Holm adjustment. The issuer filing evidence creates no new independent national adoption choices, so the entire matrix is refitted to the unchanged lagged policy input. We do not insert later liquidity estimates into earlier learning signals.

Reporting rules differ in their covered firms, audience and information dates. Liquidity associations change with both family weights and issuer scope. The evidence motivates a conditional account of disclosure effects, while pretrends and incomplete eligibility histories constrain causal conclusions. Complete source dispositions, unsupported cells and alternative specifications accompany the exhibits.

Table 1. Implemented regulation families and issuer evidence

Economic regulation family	Geographic support First / balanced	Covered-issuer evidence / remaining limit
Audit oversight / rotation	Yes / Yes	Auditor/client history needed
Expanded auditor reporting	Yes / Yes	US CAM filing cohorts
Beneficial ownership	Yes / Yes	Listed exemptions; venue gaps
Banking public CbCR	Yes / Yes	Bank perimeter; 3 dated publishers
Confidential tax CbCR	Yes / Yes	Nonpublic returns; UPE gaps
General public tax CbCR	Yes / No	No observed eligible filing cohort
Climate disclosure	Yes / Yes	Historical thresholds/report gaps
Sectoral climate reporting	No / No	Sector and panel boundary
Digital assets	Yes / Yes	Crypto object; equity scope differs
Comprehensive mandatory E/S/G	Yes / Yes	3 German statutory report publishers
Extractive payments	Yes / No	Sector/Form SD report gaps
Corporate governance	Yes / No	Listing and size phases missing
IFRS reporting	Yes / No	4 dated Canadian first IFRS reporters
Supply-chain reporting	Yes / Yes	3 US first-cycle reporters
Financial sustainability disclosure	Yes / No	Late phases; actual reports missing
Informed sustainability reporting	No / No	Scope and calendar unsupported
XBRL / digital filing	Yes / No	US early phase; first exhibit unknown

Coverage through 2025 only; future schedules are excluded. Geographic support is outcome support, not verified legal eligibility. Covered issuer studies support six families in the first year and four in the balanced observed-report subset. Country and issuer thresholds, exemptions, reporting periods and actual publication dates remain separate. Full legal scopes and event-level gap overlays are online.

Table 2. Both cross-family weighting schemes

Population / coverage / outcome	Equal family Estimate [interval]	Market-cap family Estimate [interval]
A. Geographic issuer exposure First year (15 families): Log spread	-0.049 [-0.082, -0.016]	+0.015 [-0.027, +0.056]
A. Geographic issuer exposure First year (15 families): Log price impact	-0.072 [-0.115, -0.030]	+0.009 [-0.059, +0.078]
A. Geographic issuer exposure First year (15 families): Zero-return fraction	-0.0076 [-0.0105, -0.0048]	-0.0025 [-0.0063, +0.0014]
A. Geographic issuer exposure Years 0-2 (9 families): Log spread	-0.032 [-0.088, +0.023]	+0.040 [-0.039, +0.120]
A. Geographic issuer exposure Years 0-2 (9 families): Log price impact	-0.013 [-0.080, +0.054]	+0.095 [-0.025, +0.215]
A. Geographic issuer exposure Years 0-2 (9 families): Zero-return fraction	-0.0018 [-0.0077, +0.0040]	+0.0067 [-0.0073, +0.0206]
B. Source-corroborated subset First year (6 families): Log spread	-0.090 [-0.143, -0.038]	+0.191 [+0.153, +0.228]
B. Source-corroborated subset First year (6 families): Log price impact	+0.002 [-0.110, +0.114]	+0.190 [+0.127, +0.254]
B. Source-corroborated subset First year (6 families): Zero-return fraction	+0.0011 [-0.0042, +0.0063]	+0.0058 [+0.0036, +0.0080]
B. Source-corroborated subset Years 0-2 (4 families): Log spread	-0.073 [-0.157, +0.012]	-0.081 [-0.156, -0.007]
B. Source-corroborated subset Years 0-2 (4 families): Log price impact	+0.179 [+0.051, +0.308]	+0.242 [+0.134, +0.350]
B. Source-corroborated subset Years 0-2 (4 families): Zero-return fraction	+0.0005 [-0.0065, +0.0074]	+0.0017 [-0.0036, +0.0071]

GDP weights within families. Panel A intervals condition on country sampling and the observed policy calendar; Panel B intervals describe issuer sampling only and are not causal policy confidence intervals. Zero returns are fractions. Samples are fixed within each horizon; family composition differs across horizons and populations. Equal-issuer/cap within-family alternatives and full diagnostics are online. No unsupported family is assigned a zero estimate.

Table 3. Covered-issuer first-year comparisons beyond banking

Source-corroborated study	Treated / comparison issuer-cohort observations	Log spread	Log price impact	Zero-return fraction
US auditor CAM (2019-21)	1,481 / 2,767	+0.236	+0.207	+0.0063
US early XBRL (2009)	6 / 62	-0.329	-0.417	-0.0004
US conflict minerals (2014)	3 / 70	-0.085	+0.249	+0.0071
Observed bank full reports (2015)	3 / 21	+0.089	+0.077	+0.0155
Canadian first IFRS reports (2012)	4 / 42	-0.235	+0.132	-0.0096
German statutory E/S/G (2018)	3 / 35	-0.217	-0.235	-0.0124
Alternative: bank initial phase (2014)	23 / 21	+0.035	-0.343	+0.0006

Counts are issuer-cohort observations; comparison issuers can recur across CAM cohorts. GDP is allocated over the fixed study country universe. The observed-report aggregate uses the first six rows. The last row replaces banking full reports in a separate sensitivity. Sparse, selected treated groups, industry differences, common waves and subsidiary spillovers limit interpretation; REIT comparisons are industry screened rather than legally certified noncovered. IFRS controls require current domestic-parent/US-GAAP evidence; German E/S/G controls can have covered subsidiaries. Full estimates retain balanced and alternative weights.

Source review and timing

The source audit indexes 12,326 PDF entries, preserves ten selected original reports from local archives, and reviews 402 US control filings, including 173 full submissions with hashes. Nineteen primary URL requests yield 14 saved responses and five failures with explicit alternatives. Four Canadian IFRS release days and two German report/results bundle days are corroborated; BMW retains 2018 year precision. Prior cover reviews and the 30 CAM auditor checks remain preserved. No missing filing is coded voluntary or exempt merely because it is absent.

References and primary sources

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied 2023 working paper; Journal of Accounting Research (2024), doi:10.1111/1475-679X.12548.

Koning, M., G. Mertens and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies. Supplied local article.

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Royal Bank of Canada, Toronto-Dominion, Bank of Montreal and Bank of Nova Scotia. First-quarter 2012 issuer releases and preceding restatement disclosures. RBC: rbc.com/investor-relations/_assets-custom/pdf/q112release.pdf. Other exact URLs and hashes appear in the source register.

Bank of Canada and OSFI. Chartered-bank transition notes and IFRS reporting instructions: bankofcanada.ca/rates/banking-and-financial-statistics/notes-for-tables-chartered-banks/. Historical 2011/2012 fiscal boundaries are kept separate from first actual public statements.

PCAOB AS3101; SEC Interactive Data CDI Q105.04 and Release 33-9002; SEC Conflict Minerals Release 34-67716; IRS T.D. 9773. Legal texts, original documents, failed downloads, and inherited bank/date audits are retained in the companion evidence register.

Figure 1. Six source-supported regulation families

First-year GDP-weighted issuer comparisons. Bars describe conditional issuer sampling, not causal policy confidence intervals. Every supported family appears; banking enters once. Right panel shows pre-event market-cap family weights. Zero-return outcomes are fractions. Small treated groups and different industry/country exposures remain material.

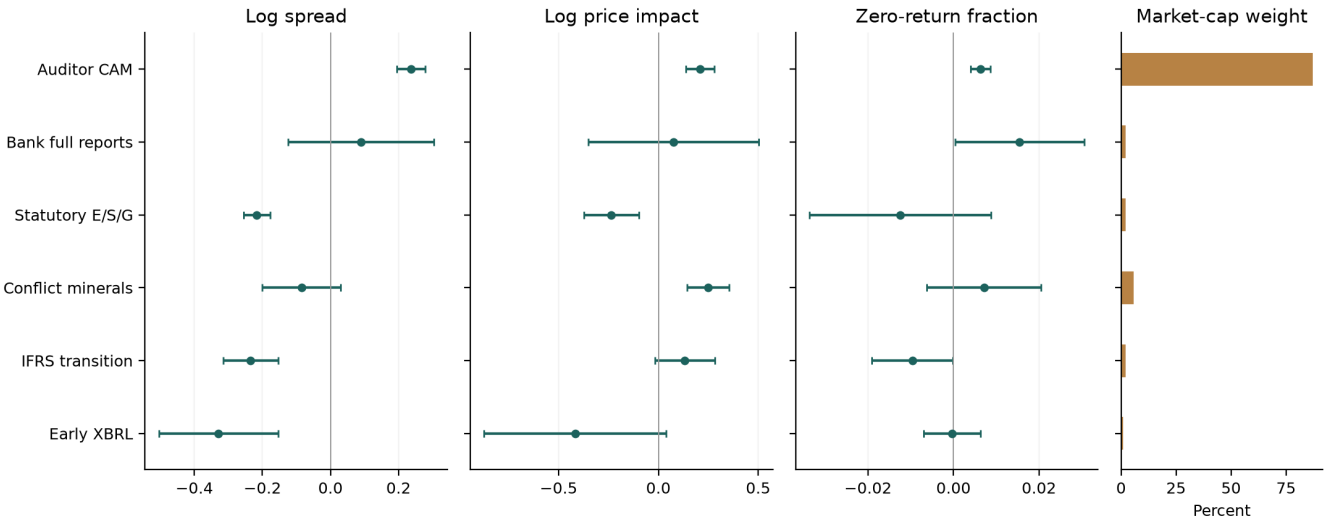


Figure 2. Four-family source-supported balanced profiles

Fixed g-3 through g+2 support: observed bank reports, conflict minerals, Canadian IFRS and German statutory nonfinancial reports. Year -1 is zero. Shading describes pointwise issuer sampling, with shared-issuer covariance retained across families. These sparse collective policy waves do not support causal policy inference. The broad geographic panel has separate adverse lead diagnostics.



Reading the exhibits

The broad exposure panel and the source-corroborated studies answer different questions. The first tracks issuer outcomes around geographic regulation calendars; the second conditions on specific legal or report evidence. The six-family first-year and four-family balanced estimates differ in both composition and horizon. A favourable geographic coefficient cannot close an eligibility gap. First-year and balanced windows also change support, and should not be joined into one event path.

Reproducible evidence

The companion report provides the frozen design, all family and aggregate estimates, weight contributions, coverage failures, alternative weights, source dispositions and 441 additive event-review JSONs. Licensed issuer and daily market data remain local. The complete diffusion matrix includes failures and coefficients opposing the hypothesised mechanisms. https://hunternbh.github.io/policy-diffusion/